FIELD GUIDE FG-001

WAFT FIELD GUIDE

Level 1: Layman's Guide (Printer Friendly)

PUBLIC

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INTRODUCTION: WHAT IS WAFT?

WAFT stands for **Wave Agent Framework & Tools**. Think of it as a laboratory where AI agents can evolve, learn, and improve themselves through a process similar to biological evolution.

> **[NOTE] SIMPLE ANALOGY**
>
> Imagine if computer programs were like living creatures. WAFT is the ecosystem where they can breed, mutate, and evolve. The strongest programs survive, and over time, they get better at their jobs.

THE BIG PICTURE: WHY THIS MATTERS

Most AI systems are static—you build them once and they stay the same. WAFT is different. It creates AI agents that can:

- **Modify their own code** (like DNA mutations)
- **Test themselves** in challenging environments
- **Evolve** by keeping what works and discarding what doesn't
- **Document themselves** automatically
- **Track their family tree** like a scientific experiment

[CAUTION] IMPORTANT CONCEPT

WAFT isn't just a tool—it's a **scientific instrument** designed to study how artificial intelligence can evolve and improve over time. The goal is to observe how AI agents develop through thousands of generations.

EQUIPMENT CHECKLIST

WHAT YOU NEED TO GET STARTED

[ ] Python 3.10 or newer installed

[ ] uv package manager (for managing dependencies)

[ ] A computer with internet connection

[ ] Basic understanding of command-line tools

[ ] Curiosity about AI and evolution

[NOTE] NOTE

You don't need to be a programmer to understand WAFT, but some technical comfort helps. This guide will explain everything in simple terms.

QUICK START: YOUR FIRST WAFT PROJECT

Step 1: Install WAFT

1. Open your terminal (command prompt on Windows, Terminal on Mac/Linux)
2. Install WAFT using: `uv tool install waft`
3. Wait for installation to complete (this may take a minute)
4. Verify installation by typing: `waft --version`

Step 2: Create Your First Laboratory

1. Choose a name for your project (e.g., "my_first_agents")
2. Run: `waft new my_first_agents`
3. WAFT will create a folder with everything you need
4. Navigate into it: `cd my_first_agents`

[WARNING] WARNING

Don't delete the `_pyrite` folder! This is WAFT's memory system. It stores all the knowledge about your project.

CORE CONCEPTS EXPLAINED SIMPLY

1. Code as DNA

In WAFT, an agent's code is like its DNA. Just as living creatures have genetic code that determines their traits, AI agents have Python code that determines their behavior.

> **[NOTE] EXAMPLE**
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> If you change an agent's code (a "mutation"), it might become better at solving problems, or it might break. WAFT tests these mutations to see which ones are improvements.

2. Evolution Through Testing

WAFT has a "gym" where agents are tested on their ability to handle errors and solve problems. Agents that perform well survive. Agents that fail are marked as "DEATH" and don't continue evolving.

3. Family Trees

Every agent has a unique ID (like a fingerprint) and WAFT tracks who "spawned" whom. This creates a family tree that scientists can study to understand how AI agents evolve.

4. Self-Documentation

One of WAFT's coolest features: it can observe its own code and write documentation about itself. This creates a "recursive loop" where WAFT improves by understanding itself better.

[CAUTION] MIND-BENDING CONCEPT

WAFT documenting WAFT using WAFT. It's like a mirror reflecting a mirror—the documentation gets better over time because WAFT learns more about itself.

COMMON QUESTIONS

Q: Do I need to be a programmer?

A: Not necessarily! This guide (Level 1) is written for anyone. However, to actually use WAFT, you'll need some comfort with command-line tools. Level 2 and Level 3 get progressively more technical.

Q: Is WAFT safe?

A: WAFT includes safety checks. Agents are tested for harmful content, errors, and dangerous behavior. Agents that fail safety tests are automatically marked as unfit.

Q: What can I do with WAFT?

A: WAFT is designed for research into AI evolution. You can:

- Create AI agents that solve specific problems
- Watch them evolve over generations
- Study how they improve
- Generate scientific data for research

Q: How long does evolution take?

A: It depends on your computer and the complexity of your agents. Simple agents might evolve in minutes. Complex ones could take hours or days. The goal is to observe evolution over thousands of generations.

SAFETY WARNINGS

[WARNING] CRITICAL WARNING

WAFT agents can modify their own code. Always test agents in isolated environments. Never run untrusted agents on systems with sensitive data. Use WAFT's safety systems and verify agent behavior before deployment.

[CAUTION] CAUTION

Evolution is unpredictable. An agent that performs well in testing might behave differently in real-world scenarios. Always validate agent behavior before using it for important tasks.

WHAT MAKES WAFT SPECIAL?

WAFT has several unique features that make it different from other AI frameworks:

Table 1: WAFT's Unique Features

Feature	What It Means	Why It Matters
Self-Modification	Agents can change their own code	Enables true evolution, not just execution
Fitness Testing	Agents are tested in a "gym"	Only the best agents survive
Lineage Tracking	Every agent's family tree is recorded	Enables scientific analysis
Self-Documentation	WAFT documents itself automatically	Creates recursive improvement loop
12 Document Templates	Professional document generation	Creates publication-ready outputs
Gamification	D&D-style progression system	Makes development engaging

NEXT STEPS

Now that you understand the basics:

1. Read **Level 2: Professional Guide** for technical details and how to actually use WAFT in your projects
2. If you're a researcher, read **Level 3: ML AI Scientist Guide** for deep scientific methodology
3. Try creating your first WAFT project using the Quick Start guide above
4. Explore WAFT's 12 document templates to see what it can generate

> **[NOTE] REMEMBER**
>
> WAFT is a scientific instrument. The goal isn't just to build AI agents—it's to **understand how AI evolves**. Every experiment contributes to our understanding of artificial cognition.

CONTACT & RESOURCES

Documentation: Check the README.md in your WAFT installation
Examples: See the `examples/` directory for code samples
GitHub: https://github.com/ctavolazzi/waft
Issues: Report problems on GitHub Issues

APPENDIX A: GLOSSARY

Table 2: Key Terms Explained

Term	Simple Explanation
Agent	An AI program that can perform tasks and modify itself
Genome	The unique code and configuration of an agent (like DNA)
Mutation	A change to an agent's code or configuration
Evolution	The process of agents improving over generations
Fitness	How well an agent performs in tests
Scint Gym	The testing environment where agents are evaluated
Flight Recorder	The system that tracks all agent actions and lineage
_pyrite	WAFT's memory system (stores project knowledge)

REMEMBER: WAFT is about understanding evolution, not just building tools.